

Nikhil Bhaskar Sardar

PhD Research Scholar

Indian Institute of Technology, Roorkee

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Research Interest: Electrical Vehicles (EVs), Artificial Intelligence/Machine Learning, Motor Control, Cyber-Security for EVs, Fast & Wireless Charging, E-bicycle Automation, Electrical Power Systems, Power Electronics

Electrical Engineering-GATE Qualified: 2014-15 & 2018-19

EDUCATION

Indian Institute of Technology, Roorkee

January 2025 (Ongoing)

PhD Research Scholar in Hydro & Renewable Energy Department

CGPA: 8.33/10

Project Title: Cyber Security in Electrical Vehicles.

Relevant Coursework: Electrical Vehicular Technology, Advanced Model Order Reduction Techniques in Control Systems, Introduction to AI/ML Tools

National Institute of Technology, Warangal

August 2015 – May 2017

Master of Technology in Electrical Power System Engineering

CGPA: 8.1/10

Scholarship: MHRD GATE Scholarship

Relevant Coursework: Power Electronics, Distribution Systems, Electronics Devices & Circuits, Electrical Vehicle Charging Methods, High Voltage AC, Control Systems, and Power Systems.

Walchand College of Engineering, Sangli

July 2009 – June 2013

Bachelor of Technology in Electrical Engineering

CGPA: 7.42/10

HSC

June 2007-2009

Maharashtra State Board

Percentage: 78

SSC

June 2007

Maharashtra State Board

Percentage: 80.15

RESEARCH EXPERIENCE

Indian Institute of Technology, Roorkee

January 2025 (Ongoing)

Supervisor: Dr. Prashant Surana, Department of HRE, IIT Roorkee

Co-Supervisor: Dr. Gaurav Chaudhary, TUD Compute, Technical University of Denmark, Denmark.

PhD Thesis Title: Cyber Security for Electrical Vehicles.

- Based on literature survey, submitted and presented an international conference paper titled “Cyber-Attack Analysis and Investigation on PMSM Drive System in Battery Electrical Vehicles” in IEEE Energy Conversion Congress & Exposition Asia (ECCE-Asia 2025)
- Submitted a paper for 2025 IEEE 4th International Conference on Smart Technologies for Power, Energy, and Control December 10-13, 2025 | NIT Goa, India with titled “Exploring Cyber Risks in Grid-Integrated Electrical Vehicle On-Board Charger (OBC)”

National Institute of Technology, Warangal

August 2015 – May 2017

Supervisor: Dr. Sydulu Maheswarapu

Master’s Thesis: Real-Time Monitoring of Points of Common Coupling in Distribution Systems through State Estimation.

- Developed a detailed model of a distribution network for real-time monitoring of Points of Common Coupling (PCC).

- Conducted a root-vector-based algorithm for observability tests on the IEEE 34 bus system, integrating pseudo measurements to enhance system observability.
- Implemented and rigorously compared the results of the forward/backward sweep algorithm and an improved load flow method for the IEEE 34 bus distribution system.
- Implemented state estimation in the distribution system using the weighted least squares method, ensuring accurate estimation of system states.
- Identified and mitigated bad data using collinearity test techniques, ensuring the integrity of the estimation process.
- Detected and mitigated unplanned reactive and active power inputs at PCC to prevent grid isolation without a system fault.
- Conducted a thorough assessment and comparison of estimated results with planned power injections to ensure the reliability of the monitoring system.
- Recognized and analyzed unscheduled power injections at PCC caused by natural calamities, providing insights into their impact on distribution networks.

Walchand College of Engineering, Sangli

July 2009 – June 2013

Supervisor: Prof. Madhukar Wavare

Bachelor's Thesis: Energy Audit at (KPT) Kulkarni Power Tools Ltd., Shirol

- Utilized a power analyzer device to measure and analyze the power factor of the company's electrical system.
- Identified inefficiencies and areas for improvement in the power factor to reduce energy consumption and improve overall efficiency.
- Designed and implemented an active filter using MATLAB to mitigate power factor issues and improve the power quality of the electrical system.

PUBLICATIONS

Conference Papers

1. Nikhil B Sardar, Prashant Surana, Gaurav Choudhary, Amit Kumar, Gianluca Gatto, "Cyber-Attack Analysis and Investigation on PMSM Drive System in Battery Electrical Vehicles", 2025 IEEE Energy Conversion Congress & Exposition Asia (ECCE-Asia)
2. Nikhil B Sardar, Prashant Surana, Gaurav Choudhary, "Exploring Cyber Risks in Grid-Integrated Electrical Vehicle On-Board Charger (OBC) ", 2025 IEEE 4th International Conference on Smart Technologies for Power, Energy, and Control December 10-13, 2025 | NIT Goa, India (**Submitted**)
3. Mohan P Thakre, Yogesh V. Mahadik, Nikhil B Sardar, "Architecture of a HV Power Battery Protection Devices for Hybrid Electric Vehicles," (HEV), - IOP Conf. Series: Materials Science and Engineering, ICCSSS 2020
4. Megha Jadhav, Aarti Chaure, Aditya Netake, Indrajit Pawar, Nikhil B Sardar, Vaishali Katkar, "Regenerative System for Mountain E-Bicycle", 1st International Conference on Circuits, Power, and Intelligent Systems (CCPIS), 2023
5. Vaishnavi Bhagwat, Harshvardhan Barge, Ganesh Naik, Vaishali Katkar, Nikhil B Sardar, Robust Estimation of State of Charge for Electric Vehicular Application", 1st International Conference on Circuits, Power, and Intelligent Systems (CCPIS), 2023
6. Mayur M. Pawar, Shubham V. Ranbhare, Shree G. Mane, Nikhil B. Sardar, "Emergency Robot", Proceedings of Cybernetics, Cognition and Machine Learning Applications (ICCCMLA 2020)
7. Renuka D. Modak, Vikramsinh A. Doke, Sayali U. Kawarkar, Nikhil B Sardar "Wireless Battery Monitoring System for Electric Vehicle", Proceedings of Cybernetics, Cognition and Machine Learning Applications (ICCCMLA 2020)
8. Patil, D. Lakade, S. Khute and N. Sardar, "Throttle Less Electric Bicycle Controller Using Fuzzy Logic," 2023 IEEE Pune Section International Conference (PuneCon), Pune, India, 2023, pp. 1-4, doi:10.1109/PuneCon58714.2023.10450099

Patent

1. “Throttle-less E-bicycle” - Application No.202221075671A has been published under Section 11A of the Indian Patents Act on January 27, 2023.

International Journal

1. Abhishek Kumar, Rushit Trivedi, Aruna Kumari, Nikhil Sardar, “Single Axis Sun Tracking Photovoltaic system using Microcontroller”, International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET), Volume 10, Issue 5, May 2021, DOI:10.15680/IJIRSET.2021.1005239

National Journal

1. Apurva Borghare, Arati Bhosale, Shital Kardile, Nikhil Sardar, “Smart Energy Monitoring using ARM Cortex ”, SAMRIDDHI – A Journal of Physical Sciences, Engineering & Technology Vol. 12, Special Issue 2, 2020, ISSN: 2454 – 5767. RNI No. UPENG04179

FUNDED PROJECT

1. Design and development of environment friendly E-bicycle: (2024-2026), Funded by SPPU ASPIRE Research Mentorship Scheme (Rs. 270k, 20TEC000818) (Ongoing)
2. Design of Developing a Controller for an Electric Scooter: (2025-2026), Funded by College of Engineering, Pune (Ongoing)

TEACHING EXPERIENCE

MIT Academy of Engineering, Alandi (D), Pune, India

August 2018– Present

Assistant Professor

- Conducted research and patented a "Throttle-less E-bicycle" to improve electric bicycle efficiency and usability.
- Researched fuzzy logic-based controllers for Brushless DC (BLDC) and Permanent Magnet DC (PMDC) motors to enhance motor control and efficiency.
- Taught courses in Electrical and Electronics Engineering, Power Electronics, Calculus and Differential Equations, and Electronics Devices and Circuits.
- Mentored students on research projects and guided them in publishing papers in reputed journals and conferences.
- Engaged in curriculum development and enhancement activities to ensure alignment with industry standards and technological advancements.

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

June 2017 – May 2018

Assistant Professor

- Taught courses in Power Systems, Power Electronics, and Electronic Devices and Circuits.
- Developed course materials and syllabi in alignment with the curriculum and industry standards.
- Conducted laboratory sessions to provide hands-on experience to students in the relevant subjects.
- Mentored students and guided them in their academic and research projects.

SKILLS

Programming: C, MATLAB Programming

Research Equipment: Multimeter, Breadboard, Power supply, oscilloscope, function generator, **Semiconductor parameters analyser.**

Software: Simulink, MATLAB

CERTIFICATIONS

Certification Name: Control Systems

NPTEL, IIT Madras

Elite with Silver, 84%

Certification Name: **Electrical Distribution System Analysis**

NPTEL, IIT Roorkee

Elite, 62%

Certification Name: **Machine Learning for Engineering and Science Applications**

NPTEL, IIT Madras

Elite, 60%

LEADERSHIP EXPERIENCE

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1. TEDxMITAOE Convener (2021, 2022, 2023)
 2. Performer at TEDxMITAOE 2020
 3. Cultural club faculty coordinator at MIT AOE, Alandi (D), Pune.
 4. Corporate Relations and Placement Cell (CRPC) faculty coordinator in the department of E&TC, MIT AOE, Alandi (D), Pune.

HOBBIES

Singing, Song Composition, Bass Guitar, Piano, Travelling

REFERENCES

Dr. Prashant Surana

Assistant Professor

Department of Hydro and Renewable

Indian Institute of Technology, Roorkee

prashant.surana@hre.iitr.ac.in

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Dr. Gaurav Choudhary

Assistant Professor

TUD Compute

Technical University of Denmark (DTU)

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Dr. Maheswarapu Sydulu

Former Dean (Administration, R and C, Faculty Welfare)

Professor (Retired)

Department of Electrical Engineering,

National Institute of Technology, Warangal

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Dr. Dagadu S. More

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